INTELLECTHON 2K25

TITLE PAGE

- **Problem Statement ID–** 25121
- **Problem Statement Title-** Smart Ambulance
- **Domain-** Health care
- **Team Name –** BRAINY BUNCHERS
- **College name –** Prince Shri Venkatestwara Padmavathy Engineering College

BRAINY BUNCHERS

SMART AMBULANCE

INTELLECTHON
A 12 HOUR HACKATHON

PROBLEM STATEMENT :

- ❖ Emergency ambulances face delays due to traffic congestion.
- ❖ Lack of real-time communication between ambulance and hospitals.
- ❖ Hospitals are often unprepared for incoming critical patients
- ❖ Delay in treatment leads to increased mortality rates.

PROPOSED SOLUTION :

To reduce ambulance delays and improve hospital coordination, we propose a simple tech-driven system using mobile apps and GPS.

- ❖ Mobile app for uploading patient vitals
- ❖ GPS tracking for live ambulance location
- ❖ Smart traffic signal control to clear ambulance path
- ❖ Dashboard for hospitals to view bed/ICU availability
- ❖ Auto-select nearest hospital based on readiness
- ❖ Instant alerts to hospital before arrival

...TARGET AUDIENCE...

- Hospitals and ER Staffs
- Ambulance and EMS Personnel
- Patients and Families
- Traffic & government authorities
- Healthcare tech developers

UNIQUENESS AND INNOVATIONS :

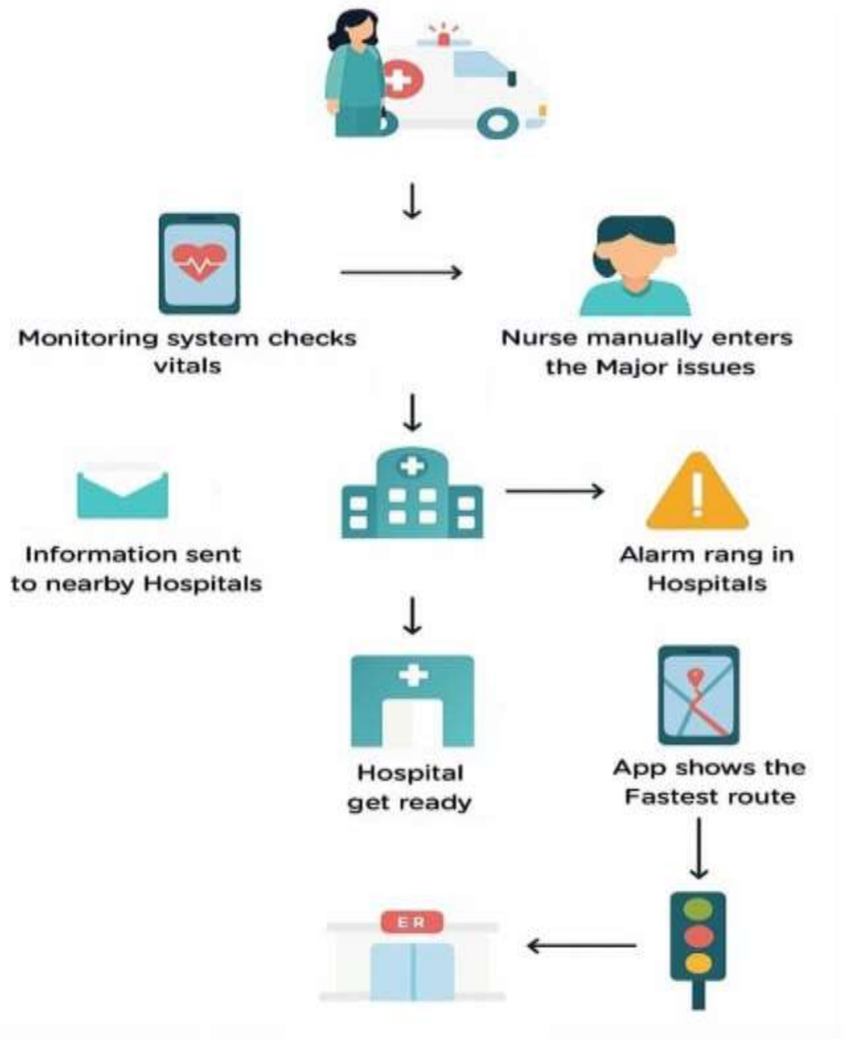
Smart Ambulance connects traffic signals, GPS, and hospitals for faster emergency care. Simple mobile tools share vitals and pick the best hospital—no delays, no heavy tech

BRAINY BUNCHERS

TECHNICAL APPROACH

INTELLECTHON
A 12 HOUR HACKATHON

WORK FLOW



Patient Enters Ambulance
↓
Vitals Monitoring System
↓
Data Sent to App
↓
Nurse enters injuries
↓
App Searches Nearby Hospitals
↓
Send Data to Hospital via Cloud
↓
Hospital Alarm Triggered
↓
Fastest Route Calculation + KM Display
↓
Traffic Signal Control (RFID + GPS)
↓
Patient's Relative Chooses Preferred Hospital
↓
Hospital Ready to Receive

TECHNOLOGIES USED:

HARDWARE :
GPS,IoT Sensor, RFID + Relays , IoT Controller(Arduino/ESP32),4 G Module

SOFTWARE :
FRONTEND : React.js
BACKEND :Node.js, MongoDB
CLOUD : Firebase,AWS

ALGORITHM :
Dijkstra's(fastest Route Algorithm)

BRAINY BUNCHERS

FEASIBILITY AND VIABILITY

INTELLECTHON
A 12 HOUR HACKATHON

TECHNICAL FEASIBILITY :

Simple tools like GPS and mobile apps

COST EFFECTIVENESS

Low-budget, open-source setup

OPERATIONAL

Easy for hospitals and ambulance teams to adopt

Potential Challenges...

- **Traffic signal access needs city approval**
- **Hospital data may not be live**
- **GPS/network issues in remote areas**
- **Scaling across different cities is tricky**

Strategies to Overcome...

- **Collaborate with local authorities**
- **Add manual update option for hospitals**
- **Use offline GPS fallback**
- **Build flexible, city-ready modules**

BRAINY BUNCHERS

IMPACTS AND BENEFITS

INTELLECTHON
A 12 HOUR HACKATHON

IMPACT :

- Faster Help: Ambulances reach patients quickly with smart routes and instant hospital alerts.
- Saves Lives: Hospitals prep in advance, reducing treatment delays.
- Bridges Gap: Connects emergency teams directly with hospitals and patients in need.

BENEFITS :

- Real-Time Data: Hospitals get patient vitals before arrival.
- Saves Time: Cuts travel and waiting time, improves critical care.
- Cost-Effective: Uses tech that's affordable and easy to expand

Scalable
Smart system can help cities, towns, and rural areas equally.

Better Access
Everyone gets emergency care, faster.

Improved Outcomes
Early data and faster transit reduce patient risks.

BRAINY BUNCHERS

RESEARCH AND REFERENCES

INTELLECTHON
A 12 HOUR HACKATHON

RESEARCH INSIGHTS :

- Emergency response times are critical; delays over 8–10 minutes can increase mortality in critical patients.
- IoT-based ambulance tracking improves route optimization and hospital preparedness.
- GPS and routing algorithms reduce travel time in urban traffic.
- Real-time patient vitals sharing enables faster and better treatment at hospitals.

REFERENCES :

P. Kaur and R. Singh, IoT-Enabled Smart Ambulance System, International Journal of Healthcare Technology, vol. 7, no. 3, pp. 45-52, 2021. [Online].
Available : https://ieeexplore.ieee.org/document/9675432